Grade 4 Science – Midterm Sheet (October)

Science · Mohamed Khair · Knowledge City Schools · Answer key ✅

Q1: Choose the correct answer

💡 Review: engineers solve problems; civil = roads and bridges; medical = devices for doctors; mechanical = materials, heat and energy; design process = identify a problem → research → prototype → test → share results.

1. People who invent and design products to help us are called:

A Doctors
B Engineers ✓
C Teachers

Why: Engineers invent and design products to help us.

2. Engineering is the process of using science and math to find solutions to:

A Games
B Problems ✓
C Stories

Why: Engineering uses science and math to solve problems.

3. Which engineer oversees the construction of roads, bridges, and skyscrapers?

A Civil engineer ✓
B Medical engineer
C Mechanical engineer

Why: Civil engineers build roads, bridges and skyscrapers.

4. Medical engineers develop devices to help:

A Build roads
B Doctors diagnose and treat diseases ✓
C Design cars

Why: Medical engineers make devices that help doctors diagnose and treat diseases.

5. Mechanical engineers study materials, heat, and:

A Plants
B Energy ✓
C Animals

Why: Mechanical engineers study materials, heat and energy.

6. What do medical engineers use to perform surgery?

A Lasers and robots ✓
B Hammers and nails
C Telescopes

Why: Medical engineers use lasers and robots in surgery.

7. Scientists use a microscope to see things that are:

A Very large
B Far away in space
C Too small to see with bare eyes ✓

Why: A microscope shows things too small to see with bare eyes.

8. Which tool helps scientists take pictures of planets and stars?

A Microscope
B Hubble Telescope ✓
C Clock

Why: The Hubble Telescope takes pictures of planets and stars.

9. Which scientist discovered that electricity causes quartz crystal to vibrate?

A Pierre Curie ✓
B Doug Engelbart
C Bill English

Why: Pierre Curie discovered that electricity makes quartz vibrate.

10. Engineers designed watches using quartz crystals to keep:

A Temperature
B Time ✓
C Distance

Why: Quartz watches keep time.

11. An electronic whiteboard in class is an example of technology in:

A Space
B Learning ✓
C Medicine

Why: A whiteboard in class is technology used for learning.

12. A satellite orbits around a planet or a:

A Star ✓
B Bridge
C Building

Why: Satellites orbit planets or stars.

13. What is the Moon?

A An artificial satellite
B A natural satellite ✓
C A robot

Why: The Moon is a natural satellite of Earth.

14. Satellites help scientists collect data about:

A Weather and land masses ✓
B Bridges and buildings
C Human heart

Why: Satellites collect data about weather and land.

15. What system uses satellites to help in navigation?

A X-ray
B GPS ✓
C Laser

Why: GPS uses satellites for navigation.

16. What do engineers use to launch satellites into space?

A Rockets ✓ B Cars

C Trains

Why: Rockets launch satellites into space.

17. To "design" means to:

A Buy products **B Make or draw plans to build something ✓**

C Destroy tools

Why: To design means to make or draw plans to build something.

18. What is the first step in the design process?

A Build a prototype B Test the prototype

C Identify a problem ✓

Why: The first step is to identify a problem.

19. After identifying a problem, engineers do:

A Research ✓ B Redesign

C Final test

Why: After identifying a problem, engineers do research.

20. Who invented the computer mouse?

A Pierre Curie **B Doug Engelbart ✓**

C Warren Morrison

Why: Doug Engelbart invented the computer mouse.

21. The first version of a test model built by engineers is called a:

A Graph **B Prototype ✓**

C Table

Why: A prototype is the first test model.

22. The first computer mouse was made of:

A Plastic **B Wood ✓**

C Glass

Why: The first computer mouse was made of wood.

23. Why do engineers test a prototype?

A To see if it meets the needs ✓ B To sell it immediately

C To throw it away

Why: Engineers test a prototype to see if it meets the needs.

24. To share test results, engineers use tables and:

A Toys
B Graphs ✓
C Stories

Why: Results are shared using tables and graphs.

25. Communicating results helps engineers find solutions to:

A Math homework
B Problems with a design ✓
C Daily meals

Why: Sharing results helps find problems with a design.

26. When engineers evaluate user reports, they may need to:

A Stop working
B Redesign the prototype ✓
C Hide the design

Why: User reports may lead to redesigning the prototype.

27. Engineers choose the best solution that costs less money and is:

A Dangerous
B Safe to use ✓
C Very heavy

Why: The best solution costs less and is safe to use.

28. Which tool is used to measure the length of a small object like a mouse?

A Ruler ✓
B Thermometer
C Scale

Why: A ruler measures the length of small objects.

29. Whiteboards enable teachers to show photos, maps, and:

A Food
B Videos ✓
C Clothes

Why: Whiteboards show photos, maps and videos.

30. Engineering is the process of using science and math to solve:

A Problems ✓
B Games
C Letters

Why: Engineering solves problems.

Q2: Complete the sentences

💡 Use the word bank: Civil, Medical, microscope, Hubble Telescope, natural, prototype, graphs, mouse, problems.

31. _____ engineers oversee the construction of highways and bridges.

A natural
B Civil ✓
C prototype
D mouse

32. **_____ engineers design X-rays, lasers, and robots for doctors.**

A natural
B Hubble Telescope
C Medical ✓
D Civil

33. **The _____ helps scientists see objects that are too small to see with bare eyes.**

A microscope ✓
B prototype
C Medical
D mouse

34. **The _____ helps scientists take accurate pictures of space.**

A natural
B Hubble Telescope ✓
C mouse
D Civil

35. **The Moon is a _____ satellite orbiting Earth.**

A natural ✓
B graphs
C Hubble Telescope
D prototype

36. **A _____ is the first working version of a solution built by engineers.**

A mouse
B Hubble Telescope
C prototype ✓
D Medical

37. **Engineers collect data and represent it using tables and _____.**

A Hubble Telescope
B graphs ✓
C microscope
D Medical

38. **Doug Engelbart invented the computer _____.**

A graphs
B mouse ✓
C Medical
D prototype

39. **Engineers use science and math to find solutions to _____.**

A graphs
B problems ✓
C Hubble Telescope
D prototype

Q3: Match

💡 GPS → navigation; microscope → small things; prototype → first test model; mechanical engineer → machines and tools; Hubble Telescope → photos of stars.

40. **Match: GPS**

A Uses satellites for navigation ✓
B Helps scientists see small things
C First test model of a design
D Takes photos of stars and galaxies

41. Match: Microscope

- **A Helps scientists see small things ✓**
- B First test model of a design
- C Takes photos of stars and galaxies
- D Designs machines and tools

42. Match: Prototype

- **A First test model of a design ✓**
- B Takes photos of stars and galaxies
- C Designs machines and tools
- D Helps scientists see small things

43. Match: Mechanical Engineer

- A Takes photos of stars and galaxies
- B Uses satellites for navigation
- C Helps scientists see small things
- **D Designs machines and tools ✓**

44. Match: Hubble Telescope

- A First test model of a design
- **B Takes photos of stars and galaxies ✓**
- C Designs machines and tools
- D Uses satellites for navigation

Q4: True or False

💡 True/False tip: one wrong word makes the whole sentence false (e.g. the Moon is natural, not artificial; a prototype is a test model, not the final product).

45. Civil engineers oversee the construction of highways, bridges, and skyscrapers.

- **A True ✓ ✓**
- B False ×

Why: Civil engineers do build highways, bridges and skyscrapers.

46. The Moon is an artificial satellite created by engineers.

- A True ✓
- **B False × ✓**

Why: The Moon is a natural satellite, not artificial.

47. Medical engineers use lasers and robots to help doctors perform surgery.

- **A True ✓ ✓**
- B False ×

Why: Medical engineers do use lasers and robots in surgery.

48. Scientists use a telescope to see things that are too small to see with bare eyes.

- A True ✓
- **B False × ✓**

Why: A microscope (not a telescope) shows very small things.

49. Doug Engelbart invented the computer mouse.

- **A True ✓ ✓**
- B False ×

Why: Doug Engelbart did invent the computer mouse.

50. **A prototype is the final product sold in stores.**

A True ✓ B **False × ✓**

Why: A prototype is the first test model, not the final product.

51. **Engineers share their test results using tables and graphs.**

A **True ✓ ✓** B False ×

Why: Engineers do share results with tables and graphs.

Q5: Short answers

💡 Microscope and Hubble Telescope are tools for research. Engineers share results to get feedback and improve the design.

52. **Which two tools help scientists in learning or research?**

A **Microscope and Hubble Telescope ✓** B Hammer and nails

C Toys and food

53. **Why do engineers share their test results with others?**

A **To find problems with the design, get feedback, and improve it so the final product is safe and works well ✓**

B To sell the prototype immediately

C To hide the design from other engineers